

GHOWARTHAN KARUNANIDHI

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EDUCATION:

ARIZONA STATE UNIVERSITY

ARIZONA, USA

Master of Science in Robotics And Autonomous Systems (Systems Engineering) 2026 Aug-2028

Relevant Coursework: Applied Linear Algebra for EGR, Robotic Systems I, Machine Learning and AI.

PANIMALAR ENGINEERING COLLEGE

CHENNAI, INDIA

Bachelor of Engineering, Mechanical Engineering,

2021-2025

GPA: 7.14/10.

Relevant Coursework: Engineering Mathematics, Mechanical Design & Simulation, Mechanics of Materials, Design and transmission systems, Professional Readiness for Innovation, Employability, Entrepreneurship, and Organizational Behavior and Ethical Practices.

WORK EXPERIENCE:

ARIZONA STATE UNIVERSITY-ARCS (Aerial, Robotics, and Control Systems) TEMPE, USA

Propulsion Engineer, Project BRIMSTONE

Aug 2026–Present

Contribute to the propulsion subsystem of a liquid-propellant VTVL lander built for the Collegiate Propulsive Landing Challenge, a national competition requiring autonomous vertical takeoff, hover, and propulsive landing. Size the propellant feed system and perform pressure and flow analyses to optimize combustion performance, working across the design-build-test cycle alongside the avionics and GNC team responsible for closed-loop vehicle control.

DOOSAN BOBCAT INDIA PRIVATE LTD

CHENNAI, INDIA

Manufacturing Engineering

Jan 2026-Aug2026

Supported shop-floor production through cycle-time monitoring, Lean/Kaizen, 5S initiatives, and process documentation. Created SWI, SOP, and MBOM; managed new product line setup, special tool purchasing, and fabrication. Designed fixtures and trolleys using SolidWorks and defined processes in Creo. Developed PFMEA documentation and contributed to Industry 4.0 and automation projects bridging robotics and AI concepts.

SRM INDUSTRIES

CHENNAI, INDIA

Manufacturing Engineering

Jan2025-Jan2026

Owned sheet metal fabrication from raw material intake to final part delivery, operating CNC machinery and optimizing cutting paths for material utilization and throughput. Enforced quality through in-process and dimensional inspections, tracing recurring failure modes to corrective actions that reduced rework, and standardized fabrication practices and safety requirements across the work cell. Partnered with the design team on DFM reviews, driving part cost reduction and production efficiency gains.

CHENNAI METRO RAIL LIMITED (CMRL)

CHENNAI, INDIA

INTERN

Jun2024-Jul2024

During my internship, I gained practical knowledge of rolling stock subsystems, train operations, and maintenance processes, enhancing my technical expertise and understanding of safety and performance standards in railway operations.

SRM INDUSTRIES INTERN

CHENNAI, INDIA
May2024-Jun2024

During my internship in a sheet metal fabrication industry, I gained hands-on experience in metal cutting, bending, and forming. I operated machinery, ensured precision, and maintained safety standards to ensure optimal performance.

National Small-scale Industry Corporation AI&MLTRANING AND INTERNSHIP

CHENNAI, INDIA
Jan2022-Jan2022

During my AI & ML internship, I developed and implemented machine learning models, focusing on data preprocessing, Model training, and evaluation.

ACADEMIC PROJECTS:

Bio-composite material For Low-Weight Engineering Applications, Jul 2024

The study investigates a bio-composite material made by reinforcing Shorea robusta epoxy with snail shell particulates, aiming to explore its potential for low-weight engineering applications. The research focuses on testing its mechanical properties to assess suitability for lightweight solutions.

RFID-Based Car Ignition System Oct 2024

The RFID-based car ignition system enhances security by requiring an RFID tag for ignition, reducing theft risk. offering convenience, and can be integrated with access control systems. The system automates ignition for smoother operations. It also provides added anti-theft protection, ensuring the car can only start with a specific RFID tag integrated with a driver's license.

STRUCTURAL ANALYSIS AND EXPERIMENTAL VALIDATION OF HYBRID COMPOSITE Apr 2025

Designed and analyzed a car body using hybrid composite materials in CATIA and ANSYS. Performed experimental validation by manufacturing and testing specimens, confirming simulation accuracy. Achieved improved strength and weight reduction compared to traditional automotive materials.

LEADERSHIP EXPERIENCE:

● PECMACT (MANAGEMENT PRACTICES AND CONSTRUCTION TECHNOLOGIES)

Led backend operations and presented research at PECMACT, showcasing leadership and teamwork skills.

● MECHANICAL SYMPOSIUM

Organized a mechanical symposium, demonstrating leadership, teamwork, and event management skills.

CERTIFICATIONS:

Google Project Management.

Business Analysis and Process Management.

MATLAB and Control Concepts.

Two-day International Workshops on IC Engines and Electric Vehicles.

Drone Workshop.

KUKA ROBOTICS WORKSHOP.

SKILLS AND INTERESTS:

Programming & Software Skills: Linux, Python, C / C++, MATLAB, LabVIEW, Excel, Ms Office365.

Robotics & Automation: ROS 2, Isaac-Sim & Isaac-Lab,Rviz, Gazebo

Design, CAD & Simulation: SolidWorks, ANSYS, Fusion 360, Creo

Manufacturing & Prototyping: EBOM TO MBOM, ECN, Process

Routing, Process maps, SOP, SWI, PFD, PFMEA, CP, FAI, 3D Printing,

New Product Implementation.

Leadership & Soft Skills: Leadership and Communication, Lateral

Thinking, Event Coordination.

